

BLACKALL, AustraliaWMO: **953510**Lat: **24.4295S**Lon: **145.4306E**Elev: **282**StdP: **97.98**Time Zone: **10.00 (E10)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.5	5.1	-10.6	1.6	19.7	-8.6	1.9	18.5	9.3	20.5	7.9	18.0	1.7	160	0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.9	39.5	20.0	38.1	19.9	36.9	19.8	24.4	30.7	23.7	30.1	23.2	29.7	3.9	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.1	18.4	26.2	22.3	17.5	25.9	21.5	16.7	25.9	75.5	30.8	72.8	30.0	70.5	29.6	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.6	6.9	DB	0.0	41.6	1.2	1.8	-0.8	42.9	-1.5	44.0	-2.2	45.0	-3.0	46.4	(4)
(5)			WB	-1.7	25.2	1.3	0.9	-2.6	25.9	-3.4	26.4	-4.2	26.9	-5.2	27.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.8	29.2	28.1	26.9	23.1	18.5	15.4	14.9	16.5	20.8	24.5	27.1	28.7
	DBStd	5.99	2.69	2.60	2.60	2.75	3.06	3.06	3.05	3.32	3.52	3.38	2.94	2.95
	HDD10.0	2	0	0	0	0	0	1	1	1	0	0	0	0
	HDD18.3	339	0	0	0	2	34	96	112	76	16	3	0	0
	CDD10.0	4671	597	508	523	394	264	163	154	203	323	450	513	579
	CDD18.3	1967	338	275	265	146	40	9	7	20	88	194	263	321
	CDH23.3	24087	4462	3426	3096	1486	382	76	76	289	1118	2308	3221	4146
CDH26.7	12442	2559	1867	1568	574	68	4	6	62	452	1154	1758	2369	
Wind	WSAvg	3.4	3.6	3.5	3.4	3.1	3.0	2.9	3.0	3.2	3.4	3.7	3.8	3.7
Precipitation	PrecAvg	514	90	71	56	35	36	19	21	18	16	28	50	68
	PrecMax	787	395	230	368	299	240	98	126	119	64	121	204	247
	PrecMin	259	8	6	0	0	0	0	0	0	0	0	0	2
	PrecStd	160	83	55	70	52	44	24	31	23	20	23	47	53
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.9	40.8	38.6	34.5	30.7	27.2	27.5	30.9	35.8	38.1	39.8	41.0
		MCWB	20.3	20.4	20.4	20.3	17.6	16.3	16.3	15.5	16.8	17.9	19.3	19.6
	2%	DB	39.0	38.3	36.6	32.8	28.5	25.5	25.4	28.3	33.0	35.9	37.9	39.1
		MCWB	20.6	20.3	20.4	18.8	16.8	15.7	14.9	15.0	16.2	17.5	18.8	19.8
	5%	DB	37.6	36.4	34.9	31.3	27.0	24.1	23.9	26.5	31.2	34.3	36.2	37.5
		MCWB	20.8	20.7	20.2	18.0	16.0	15.0	14.1	14.2	15.8	17.4	18.8	19.5
	10%	DB	36.0	34.7	33.4	30.0	25.6	22.4	22.3	24.7	29.4	32.5	34.5	35.8
		MCWB	20.7	20.8	20.0	17.5	15.4	14.0	13.3	13.3	15.4	16.9	18.4	19.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.4	25.2	24.6	22.3	20.7	18.2	19.1	18.3	20.7	21.5	23.2	24.7
		MCDB	32.2	31.5	28.9	28.8	24.8	24.2	23.4	25.9	27.0	30.2	29.0	31.3
	2%	WB	24.4	24.3	23.2	21.3	18.9	17.1	17.3	16.7	19.3	20.4	21.9	23.6
		MCDB	31.0	30.3	28.7	28.4	24.8	22.2	21.8	24.7	25.7	28.9	28.7	31.0
	5%	WB	23.7	23.7	22.6	20.4	17.8	16.2	15.9	15.5	18.0	19.7	21.2	22.8
		MCDB	30.3	29.7	28.9	27.5	24.4	21.4	21.2	23.7	25.7	28.4	29.1	29.9
	10%	WB	23.1	23.1	21.9	19.4	16.8	15.3	14.5	14.4	16.9	18.9	20.5	22.2
		MCDB	29.9	29.7	29.4	26.5	23.5	20.6	20.6	22.6	26.4	27.6	29.1	29.6
Mean Daily Temperature Range	5% DB	MDBR	11.9	11.9	12.2	13.3	13.7	13.4	14.4	15.7	15.6	14.5	13.0	12.2
		MCDBR	14.0	14.3	13.9	14.4	14.6	14.3	15.9	17.1	17.4	16.2	15.2	14.5
		MCWBR	4.4	4.2	4.4	5.5	6.3	6.7	7.8	8.0	7.1	6.1	4.9	4.8
	5% WB	MCDBR	10.7	10.5	10.3	11.6	10.9	11.0	11.9	14.1	13.0	12.8	11.5	11.1
	MCWBR	3.8	3.8	3.6	4.7	5.2	5.3	6.3	6.9	5.8	5.2	4.1	4.1	
Clear-Sky Solar Irradiance	taub	0.373	0.362	0.342	0.327	0.309	0.297	0.290	0.306	0.344	0.361	0.377	0.379	
	taud	2.505	2.541	2.581	2.575	2.570	2.604	2.602	2.529	2.432	2.417	2.420	2.454	
	Ebn at Noon	970	967	959	930	908	900	921	939	941	956	960	966	
	Edh at Noon	115	109	99	92	84	78	80	94	113	121	124	121	
All-Sky Solar Radiation	RadAvg	7.28	6.96	6.26	5.39	4.41	3.82	4.22	5.23	6.20	7.12	7.36	7.43	
	RadStd	0.48	0.48	0.41	0.37	0.21	0.25	0.28	0.27	0.43	0.50	0.56	0.54	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page